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## Experiment-21

### Question

Write a menu-driven C program that accepts two positive integers,  $m$  and  $n$ , where  $m \leq n$ . Display the following menu and use a `switch` statement to perform the operation selected by the user:

1. Print the multiplication tables of  $m$  and  $n$  side by side, using multipliers from 1 to 10.
2. Display a multiplication grid containing the products of integers from 1 to  $m$  and integers from 1 to  $n$ .
3. For each integer from  $m$  to  $n$ , print that integer followed by the next five consecutive integers.
4. Exit the program.

Continue displaying the menu until the user selects Exit. Display an appropriate error message for an invalid menu choice.

### Algorithm

1. Start.
2. Declare integer variables  $m$ ,  $n$ , `choice`,  $i$ ,  $j$  and `num`.
3. Input the values of  $m$  and  $n$ .
4. Check whether  $m$  and  $n$  are positive and  $m \leq n$ .
5. If the values are invalid, display an error message and ask the user to enter them again.
6. Display the menu:
  - Multiplication tables of  $m$  and  $n$
  - Multiplication grid
  - Consecutive integers
  - Exit
7. Input the user's choice.
8. Use a `switch` statement to perform the selected operation.
9. For **Choice 1**, use a loop from 1 to 10 and print the multiplication tables of  $m$  and  $n$  side by side.
10. For **Choice 2**, use nested loops. The outer loop runs from 1 to  $m$  and the inner loop runs from 1 to  $n$ . Display the product  $i * j$  to form the multiplication grid.
11. For **Choice 3**, use an outer loop from  $m$  to  $n$ . For each number, use another loop to display that number and the next five consecutive integers.
12. For **Choice 4**, display an exit message and terminate the program.

13. For any other choice, display `Invalid choice`.
14. Repeat the menu until the user selects Choice 4.
15. Stop.

## Test Case Table

| Test Case | Inputs             | Expected Output                                           | Results |
|-----------|--------------------|-----------------------------------------------------------|---------|
| 1         | m=3, n=5, Choice 1 | Tables of 3 and 5 displayed from 1 to 10                  | Pass    |
| 2         | m=3, n=5, Choice 2 | 3 × 5 multiplication grid displayed                       | Pass    |
| 3         | m=3, n=5, Choice 3 | Consecutive integers from 3 through 5 displayed correctly | Pass    |
| 4         | m=3, n=5, Choice 4 | Program exited successfully                               | Pass    |
| 5         | m=5, n=3           | Invalid-input message displayed                           | Pass    |
| 6         | m=0, n=5           | Invalid-input message displayed                           | Pass    |
| 7         | m=1, n=1, Choice 1 | Multiplication table of 1 displayed in both columns       | Pass    |
| 8         | m=3, n=5, Choice 7 | Invalid-choice message displayed                          | Pass    |

## C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int m, n, choice;
```

```
    int i, j, num;
```

```
    do
```

```
    {
```

```
        printf("Enter m and n (positive integers, m <= n): ");
```

```
        scanf("%d %d", &m, &n);
```

```
        if (m <= 0 || n <= 0 || m > n)
```

```
        {
```

```
            printf("Invalid input. Enter positive integers with m <= n.\n");
```

```
        }
```

```

} while (m <= 0 || n <= 0 || m > n);

do
{
    printf("\n1. Multiplication tables of m and n\n");
    printf("2. Multiplication grid\n");
    printf("3. Consecutive integers\n");
    printf("4. Exit\n");

    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n\n",
                m, n, m, n);

            for (i = 1; i <= 10; i++)
            {
                printf("%d x %d = %-5d\t%d x %d = %d\n",
                    m, i, m * i, n, i, n * i);
            }
            break;

        case 2:
            printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);

            printf("    ");
            for (j = 1; j <= n; j++)
            {
                printf("%4d", j);
            }

            printf("\n    ");
            for (j = 1; j <= n; j++)
            {
                printf("----");
            }
            printf("\n");

            for (i = 1; i <= m; i++)
            {
                printf("%2d | ", i);

                for (j = 1; j <= n; j++)
                {
                    printf("%4d", i * j);
                }
            }
        }
    }

```

```

        printf("\n");
    }
    break;

case 3:
    printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n\n",
        m, n, m, n);

    for (num = m; num <= n; num++)
    {
        printf("%d: ", num);

        for (j = 0; j <= 5; j++)
        {
            printf("%d ", num + j);
        }

        printf("\n");
    }
    break;

case 4:
    printf("\nExiting program.\n");
    break;

default:
    printf("\nInvalid choice. Please enter 1 to 4.\n");
}

} while (choice != 4);

return 0;
}

```

## Compilation and Execution

```

(kali㉿kali)-[~]
$ vi hdoc22.c

(kali㉿kali)-[~]
$ gcc hdoc22.c -o hdoc22

(kali㉿kali)-[~]
$ ./hdoc22
Enter m and n (positive integers, m ≤ n): 3 5

```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 1

Multiplication Tables of 3 and 5 (m=3, n=5)

|             |             |
|-------------|-------------|
| 3 x 1 = 3   | 5 x 1 = 5   |
| 3 x 2 = 6   | 5 x 2 = 10  |
| 3 x 3 = 9   | 5 x 3 = 15  |
| 3 x 4 = 12  | 5 x 4 = 20  |
| 3 x 5 = 15  | 5 x 5 = 25  |
| 3 x 6 = 18  | 5 x 6 = 30  |
| 3 x 7 = 21  | 5 x 7 = 35  |
| 3 x 8 = 24  | 5 x 8 = 40  |
| 3 x 9 = 27  | 5 x 9 = 45  |
| 3 x 10 = 30 | 5 x 10 = 50 |

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 2

Multiplication Grid (m=3, n=5)

|   | 1 | 2 | 3 | 4  | 5  |
|---|---|---|---|----|----|
| 1 | 1 | 2 | 3 | 4  | 5  |
| 2 | 2 | 4 | 6 | 8  | 10 |
| 3 | 3 | 6 | 9 | 12 | 15 |

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

```

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

```

1. Multiplication tables of m and n

2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 4

Exiting program.

```
(kali㉿kali)-[~]  
$ gcc hdoc22.c -o hdoc22
```

```
(kali㉿kali)-[~]  
$ ./hdoc22
```

Enter m and n (positive integers,  $m \leq n$ ): 5 3  
Invalid input. Enter positive integers with  $m \leq n$ .  
Enter m and n (positive integers,  $m \leq n$ ): 0 5  
Invalid input. Enter positive integers with  $m \leq n$ .  
Enter m and n (positive integers,  $m \leq n$ ): 1 1

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 1

Multiplication Tables of 1 and 1 ( $m=1$ ,  $n=1$ )

1 x 1 = 1          1 x 1 = 1

|             |             |
|-------------|-------------|
| 1 x 2 = 2   | 1 x 2 = 2   |
| 1 x 3 = 3   | 1 x 3 = 3   |
| 1 x 4 = 4   | 1 x 4 = 4   |
| 1 x 5 = 5   | 1 x 5 = 5   |
| 1 x 6 = 6   | 1 x 6 = 6   |
| 1 x 7 = 7   | 1 x 7 = 7   |
| 1 x 8 = 8   | 1 x 8 = 8   |
| 1 x 9 = 9   | 1 x 9 = 9   |
| 1 x 10 = 10 | 1 x 10 = 10 |

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 7

Invalid choice. Please enter 1 to 4.